Design of a reausable High-Altitude Radiation Measurement System

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Abstract

Understanding atmospheric radiation is crucial for assessing both natural and anthropogenic impacts on Earth's environment. This study employs high-altitude balloons to measure radiation levels at various altitudes, providing valuable data for atmospheric research and environmental monitoring. Our system, equipped with a radiation monitor, environmental sensors, and telemetry systems, enables real-time data collection and transmission during ascent. Supported by TÜBA, TÜBİTAK BİDEB, the General Directorate of Meteorology, and Meteosis, the Test-Platform with Meteorology Balloons conducts tests of cube satellites and components up to 32 km altitude. This paper demonstrates the cost-efficiency and versatility of high-altitude balloons for atmospheric radiation studies, presenting data collected during the mission. These findings are essential for improving our understanding of high-altitude radiation environments, vital for designing and operating high-altitude aircraft and space missions.

Introduction

- The Regener-Pfotzer maximum is a peak in the intensity of secondary cosmic radiation occurring at an altitude of is influenced by several factors, including geomagnetic vertical cut-of rigidity, solar cycle phases, and particle types. Understanding these dependencies is crucial for accurately assessing radiation exposure in the upper atmosphere, which is particularly important for high-altitude flights and space missions.[1]

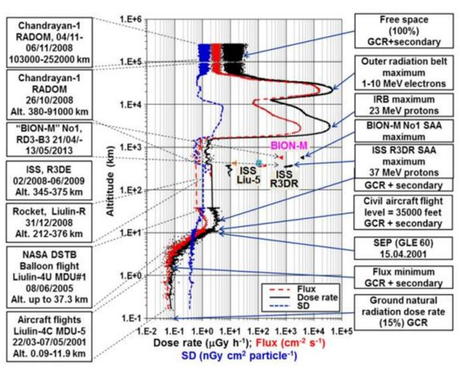


Figure 1: Variations of Absorbed Dose Rate and Flux with Altitude [2]

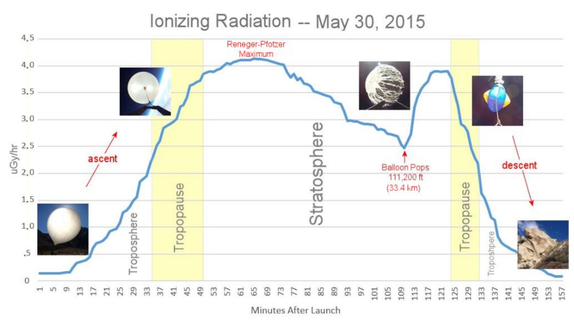


Figure 2: Dose Rate vs Time graph and The Regener-Pfotzer maximum. [22]

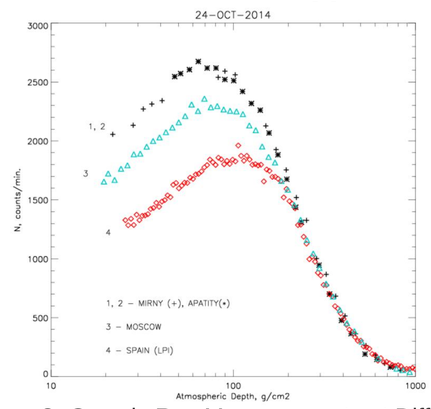


Figure 3: Cosmic Ray Measurements at Different Geomagnetic Vertical Cut-Off Rigidities [15]

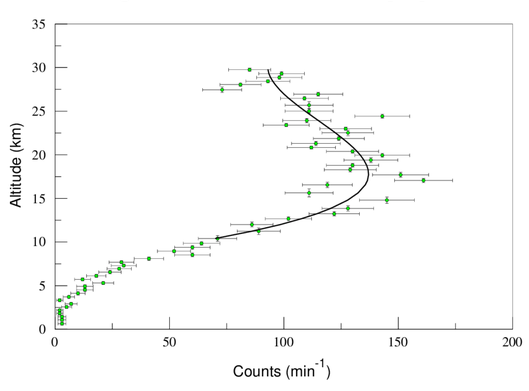


Figure 4: Altitude versus omnidirectional counts of particles during Solar cycle. [16]

- High-altitude balloons offer a low-cost, efficient, and reusable solution for measuring radiation levels over extended periods and at various altitudes. Unlike single-use sounding rockets, these balloons can carry instruments to high altitudes and maintain lower acceleration, allowing for more comprehensive data collection throughout the flight

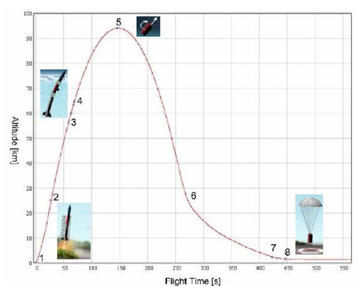


Figure 5: Altitude vs. Time Graph for Sounding Rockets [28]

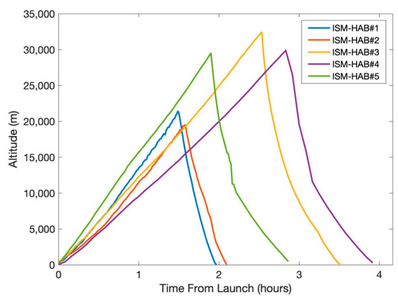


Figure 6: Altitude vs. Time Graph for Different High-Altitude Balloons [11]

Method

- In this mission, there are two electronic systems: one ground station and one payload for the balloon. The ground station is responsible for receiving real-time data, while the payload is tasked with measuring radiation and other parameters.

Electronic Components of Payload:

1. ESP32 S3 Microcontroller: Manages the mission flow.
2. SD Card: Stores data collected during the mission, accessible after landing.
3. Temperature and Pressure Sensors: Measure atmospheric conditions.
4. GPS/GNSS Module: Determines the position of the payload.

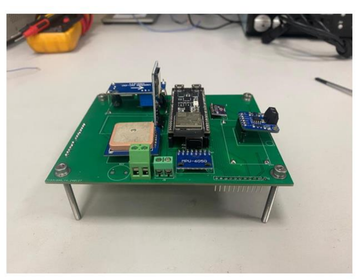


Figure 7: Production of the Payload's Printed Circuit Board (PCB), Front Views

Electronic Components of Payload:

5. LoRa Radio Communication Module: Facilitates real-time data transmission.
6. GPRS Module: Sends SMS notifications after landing.
7. Geiger Sensor: Measures radiation levels, designed by METU IVMER.
8. Buzzer and LED: Assist in locating and rescuing the payload after landing.
9. SD Card Heater: Ensures SD card functionality in extremely cold temperatures.

Mechanical Properties of The Payload:

1. Case Material: EPS Polystyrene for lightweight and heat insulation (-70°C).
2. Conical Nose Shape: Enhances stability and aerodynamics.
3. Parachute: 2m diameter for controlled descent at 3.5 m/s.
4. Payload Weight: 1100g (excluding the balloon).
5. Chassis: Secures electronics and facilitates wiring.
6. System Connection: 1600g hydrogen balloon connected to parachute with 4m rope, and parachute connected to payload with 22m rope, using a swivel for attachment.

Electronic Components of the Ground Station:

1. Microcontroller: ESP32 WROOM32 Manages the data processing
2. Receiver: LoRa SX1276 radio communication modüle Facilitates real-time data transmission.
3. Data Storage: SD card for recording real-time data

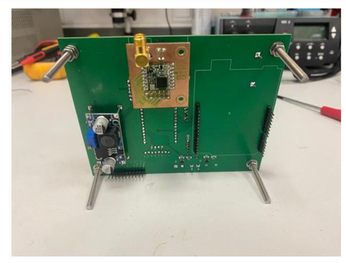


Figure 8: Production of the Payload's Printed Circuit Board (PCB), Back Views

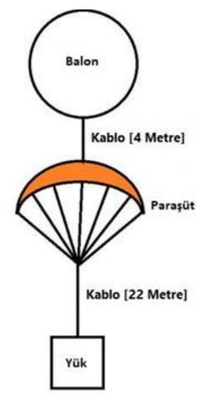


Figure 9: Mechanical Design; Flight Train

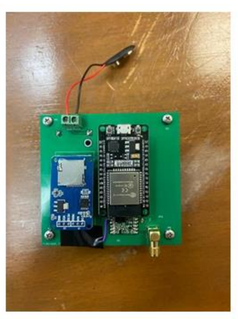


Figure 10: Production of the Ground Station's Printed Circuit Board (PCB).

- The ground station uses a directional antenna, while the payload has an omnidirectional antenna. Calculations show a 50dB gain, which is significantly higher compared to the typical 3dB gain in satellite communications. This substantial gain indicates a strong and efficient communication link between the ground station and the payload.

Results & Conclusion





Figure 11: The System Before Flight (left) and After Recovery (right)

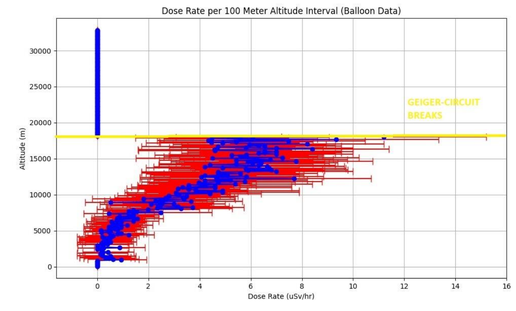


Figure 12: Mean Geiger Counter Counts per 100 Meter Altitude Interval Indicating the Regener-Pfotzer Maximum

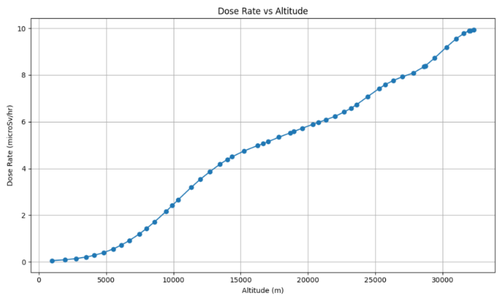


Figure 13: Average Dose Rate for 27/06/2024 at METU Calculated by CARI7-A.

- Although the Geiger circuit ceased to function due to the low resistivity of the air at decreasing pressures (the Geiger counter operates at 570V), Geiger data was successfully recorded up to an altitude of 18 km. This altitude is sufficient to observe the Regener-Pfotzer maximum. The output data is presented in Figure 9.
- The recorded data fits well with the predictions and the graph created from the CARI-7A model.
- The recovery functions operated flawlessly. Messages sent from the top of a tree 147 km away from the launch location were successfully received at a frequency of one message per minute via GPRS.
- The maximum range of the balloon was 180 km, and even at this distance, 83% of the transmitted packets were successfully received without issues.

Acknowledgements

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REFERANCES